Response to Reviewers — ARRAY-D-26-04840

Manuscript title: User Next Click Prediction Using Machine Learning for UX Analysis

Dear Editor and Reviewers,

Thank you for your careful assessment and constructive recommendations. Our aim in this revision is to address each comment through practical improvements to the manuscript, its underlying idea, and the proposed approach. We use the reviewers' recommendations to identify additional analyses, methodological checks, clarifications, and revisions that can strengthen the work. For each point, we distinguish actions completed and supported by evidence from work that remains outstanding, and explain the next step and any information or external assessment needed to complete it. Where a requested study or verification has not been performed, we acknowledge the gap and qualify the corresponding claims.

The revision corrects the train/test protocol, distinguishes the actual click-only model inputs from the inaccurate feature description, adds newly executed same-data comparisons and uncertainty estimates, and withdraws untested UX-improvement and cross-dataset superiority claims. The latest repeat-inclusive analysis adds UNCP 5, an adaptive probability fusion of UNCP 2 and a transition model. Its gate was selected on validation users; on 10,833 targets from 11 held-out users, mean Top-1/Top-3 over seeds 11, 23, and 37 is 66.71%/85.09% for UNCP 5, 66.60%/84.08% for newly executed repeat-aware UNCP 6, and 64.62%/83.58% for UNCP 2. UNCP 5 remains the highest observed model. The paired Top-3 gain is 1.51 points, with an exploratory user-cluster bootstrap interval of 1.03–1.77 points. These predictions can assist UX researchers in identifying opportunities to improve the user experience. The point-by-point responses identify the changes and supporting evidence, together with unresolved issues. Reviewer 1's unnumbered substantive comments are numbered 1–15 here for reference; Reviewers 2 and 4 retain their original numbering.

Reviewer 1

Comment 1.1 — Train/test splitting

> *The train/test splitting strategy requires substantial clarification. The manuscript states that 4,216 click sequences were divided into 80% training and 20% testing data, yielding 843 test sequences. It is not clear whether sequences from the same user can appear in both training and test sets. Since only 50 users are included, a random sequence-level split could create substantial user-identity leakage and overestimate generalization. The authors should report whether splitting was performed at the user level, session level, or sequence level. A user-independent evaluation, such as leave-user- out or group-based splitting, would provide much stronger evidence of generalizability to unseen users.*

Response Thank you for identifying this important issue. Your concern is correct: the original random split did not enforce user independence, so its scores cannot establish generalization to unseen users. More precisely, the supplied implementation expanded 4,216 source sequences into 31,074 overlapping prefix–target examples before randomly splitting them into 24,859 training and 6,215 test examples; it did not produce 843 independent test sequences. Reconstructing this procedure showed that 6,061 test examples (97.52%) shared their source sequence with training. Although user ID was not a predictive input, overlapping users and dependent prefixes make this evaluation unsuitable for the claimed unseen-user setting. This overlap demonstrates a protocol problem, not a numerical estimate of how much accuracy was inflated.

To address this issue, the revised evaluation assigns complete recorded user IDs to mutually exclusive training, validation, and test sets before generating prefix–target examples. The supplied archive contains 52 distinct user IDs, rather than the 50 users stated previously; the relationship between these identifiers and distinct participants still requires source reconciliation. The 46,194 figure counts click events, whereas the split counts are supervised next-click targets. Each sequence of length L contributes L-1 targets; therefore, 46,194 events across 4,223 sequences yield 41,971 targets. With split seed 20261103, the repeat-inclusive primary partitions contain 32/9/11 user IDs and 25,399/5,739/10,833 targets, respectively. The totals reconcile exactly: 32 + 9 + 11 = 52 User IDs and 25,399 + 5,739 + 10,833 = 41,971 targets. No user ID appears in more than one partition. Vocabularies and frequency estimates are fitted on training data only, and validation data govern early stopping. Six automated protocol tests, including pairwise user separation and training-only vocabulary/count checks, pass.

The completed user-disjoint runs on the repeat-inclusive dataset yield mean Top-1/Top-3 accuracies of 54.52%/78.00% for LSTM, 44.46%/71.92% for Transformer, and 62.72%/82.30% for UNCP, across training seeds 11, 23, and 37 on the same held-out users. The Markov-4 back-off baseline achieves 60.27%/79.91% on that test set and is retained for a transparent comparison. The user-cluster 95% bootstrap interval for UNCP Top-3 is 79.13%–84.28%; its 2.39-point advantage over Markov-4 is supported by the paired analysis (Holm-adjusted user-macro p=0.0293).

We therefore replace the original scores as evidence of unseen-user performance with the user-disjoint evaluation. This is one fixed group-based holdout with repeated model-training seeds, not leave-one-user-out cross-validation or three independent user splits. With only 11 held-out user IDs from one application, the findings support a limited within-application assessment of previously unseen IDs, not broad generalization across populations or applications. The split manifest, protocol checks, per-seed outputs, and aggregate metrics are provided to make the evaluation inspectable.

Revision in the manuscript — p. 15, Data Preparation

Manuscript text: “The primary test holds out complete User IDs. We assigned all sequences of each recorded account to exactly one of the training, validation, or test partitions before creating overlapping click-prefix examples.”

Evidence: Split manifest and automated protocol checks.

Comment 1.2 — Appropriate baselines

> *The reported model accuracy should be interpreted relative to appropriate naïve baselines. The application contains 65 clickable elements across 15 pages, with a highly variable number of clickable elements per page. In this setting, Top-3 accuracy can be inherently high on pages containing only a few possible clickable elements. For example, the manuscript itself notes that some pages contain only one clickable element. The evaluation should therefore include simple but informative baselines such as: most-frequent-click per page, previous-click/transition-frequency baseline, first-order Markov model, random prediction constrained to clickable elements on the current page. Without these comparisons, it is difficult to determine how much predictive value is contributed specifically by the neural architectures.*

Response We agree that Top-3 accuracy alone is insufficient, particularly when a page has few eligible elements. For a verified candidate set of size K containing the target, uniform sampling without replacement has expected Top-3 accuracy min(3,K)/K; thus it is 100% when K is at most three. We now explicitly separate deployable history-based baselines, page proxies, and retrospective destination-page diagnostics, which use the page of the next click and cannot be used for prediction before that click occurs.

On the current repeat-inclusive user-disjoint test set (10,833 examples from 11 held-out IDs), global popularity reaches 5.16%/22.37% Top-1/Top-3, repeating the last click reaches 14.23%/16.74%, and first-order Markov reaches 36.36%/64.58%. The stronger Markov-4 baseline reaches 60.27%/79.91%; LSTM, Transformer, and the original UNCP reach 54.52%/78.00%, 44.46%/71.92%, and 62.72%/82.30%, respectively. Graph-aware UNCP 2 reaches 64.62%/83.58%, and validation-selected UNCP 5 reaches 66.71%/85.09% over the same three seeds. All transition frequencies and the UNCP 5 smoothing prior are estimated from training data only.

To address the page-specific request as far as the available data permit, we additionally implemented and executed two explicitly labelled proxies. Ranking training-observed elements of the last logged click's page by their training frequencies achieves 13.55%/35.04%. Uniform sampling from that same training-observed candidate set has exact expected accuracy 3.52%/10.48%; this expectation is computed analytically rather than estimated from random draws. Empty candidate sets and targets outside the set score zero. A separate next-click-frequency baseline conditioned on the previous logged page, without restricting the predicted next click to that page, achieves 16.56%/40.39%. These are different, explicitly defined tasks, and none is presented as a verified current-page-constrained evaluation.

The limitation is material: the export contains 19 distinct page strings, and the stated inventory of 65 clickable elements across 15 pages has not been reconciled with an authoritative application inventory. A click's recorded page does not establish the displayed page after navigation. The next event's recorded page differs from the previous event's page in 40.86% of test examples. Using the next click's page to choose candidates would therefore introduce future information. For diagnostic purposes only, destination-page popularity and destination-page uniform sampling yield 27.66%/60.88% and 12.19%/25.66%, respectively; both use the future page and are excluded from claims about real-time prediction. These calculations use training-observed elements, not an independently verified list of all clickable elements.

The updated analysis exports aggregate and per-page results, training-observed candidate counts, and UNCP page-level metrics. Five new scoring tests cover one, three, and four candidates, unseen/off-page targets, and empty candidate sets; all five and the six existing protocol tests pass. Fully satisfying the current-page part of the request still requires the page visible immediately before each prediction and an authoritative page-to-element inventory. Once these are supplied, the same fixed user split can be used to evaluate the exact requested page-constrained baselines without substituting destination-page knowledge. We report this outstanding requirement rather than claiming the proxy analysis resolves it.

Revision in the manuscript — pp. 15–16, Quantitative Evaluation Results / Table 1

Manuscript text: "The same held-out users and 10,833 prediction targets are used for every method in Table 1."

Evidence: Table 1 and supplementary model-result files.

Comment 1.3 — Per-page performance and imbalance

> *Per-page and class-imbalance analysis is needed. The frequency of user interactions appears to vary considerably across pages and UI elements. Overall Top-1 and Top-3 accuracy may therefore be dominated by frequently visited pages or highly frequent elements. Please report performance by page, by element-frequency group, and preferably macro-averaged metrics in addition to global accuracy. This would clarify whether the models perform consistently or primarily learn dominant interaction patterns.*

Response We have now incorporated the supplied page/element inventory and extended the imbalance analysis. The inventory contains 65 explicitly named elements across 15 pages. After case-insensitive exact matching, 29,708 of 35,290 event rows (84.18%) receive a page label. We leave 5,582 rows blank because 33 labels (for example label0–label9, user0–user9, surveysBox0–surveysBox4 and several dynamic strings) do not occur in the supplied inventory; these rows are listed in page_mapping_audit.json and were not guessed. The resulting file events_source_with_page.csv preserves every original row and adds page_from_supplied_inventory.

The repeat-inclusive held-out-user analysis reports global and macro scores separately. For UNCP 2, global Top-1/Top-3 are 64.62%/83.58%, while target-label macro Top-1/Top-3 are 55.67%/71.03%; its user-macro Top-1/Top-3 are 65.47%/84.17%. UNCP 5 raises global Top-1/Top-3 to 66.71%/85.09%. The frequency-group comparison supplied with UNCP 5 shows Top-3 of 41.32% for rare, 65.49% for medium-frequency, and 87.15% for frequent targets; the aggregate score is influenced by frequent interactions. Page diagnostics remain descriptive because the recorded destination page is not an inference-time input.

We group repeat-inclusive test targets by training frequency: unseen (2 examples), rare (96), medium (822), and frequent (9,913). UNCP 5 Top-3 is 0.00%, 41.32%, 65.49%, and 87.15% in those groups, respectively; UNCP 2 reaches 0.00%, 40.63%, 62.90%, and 85.73%. This gradient shows that aggregate accuracy is influenced by frequent interactions, although the adaptive fusion improves the observed score in each known-label group.

Per-page metrics, candidate counts, per-label metrics, frequency-group metrics, and macro summaries are exported. The page labels are used for stratification and auditing only; they are not supplied as future-page inputs to the primary model. The supplied inventory resolves the previously missing 65-element/15-page reference for most rows, but the unmatched dynamic labels and the absence of a verified page-visible-at-

prediction field remain limitations. We have therefore reported both exact mapped coverage and the unresolved rows instead of claiming complete page reconciliation.

Revision in the manuscript — pp. 13–17, Evaluation Metrics and Results

Manuscript text: "In addition to micro Top-1 and Top-3 accuracy, we report macro averages across held-out User IDs and target labels, results by recorded page and by training-label frequency group, and candidate-set diagnostics."

Evidence: Page and element-frequency diagnostics.

Comment 1.4 — Cross-dataset comparisons

> *Comparisons with prior published models should be made cautiously. The manuscript compares its Top-3 accuracy of 0.88 with values reported by Zhou and Li and concludes that the proposed approach demonstrates superior predictive power. However, the datasets, application structures, number of users, candidate click spaces, preprocessing procedures, and evaluation protocols appear to differ. Direct numerical comparison across different datasets does not establish model superiority. The claim should either be softened or supported through reimplementation and evaluation of the compared methods on the same dataset and split.*

Response We agree with this comment. A numerical comparison between our former Top-3 value (0.88) and a value reported by Zhou and Li was not a controlled comparison because the datasets, users, interface structure, candidate space, preprocessing, and split/evaluation protocol differ. We therefore removed the cross-dataset superiority statement and no longer use the former 0.88 value as evidence that our model is better than a published model.

We did not claim to have reimplemented Zhou and Li without their complete executable implementation, data, candidate-space definition, and preprocessing details. Instead, we compare methods on the same repeat-inclusive data and held-out-user split. We also include naïve and conventional baselines—uniform expected prediction, global click popularity, logistic regression, RNN, GRU, and Markov-4—and report them alongside LSTM, Transformer, and the UNCP variants in Table 1 under the same held-out-user evaluation. Top-3 is 79.91% for Markov-4, 78.00% for LSTM, 78.88% for GRU, 71.92% for Transformer, 82.30% for UNCP, 83.58% for UNCP 2, 84.08% for UNCP 6, and 85.09% for UNCP 5. These within-dataset comparisons do not establish superiority over published results on different datasets. The UNCP 5 gain over UNCP 2 is 1.51 percentage points, with an exploratory paired held-out-user bootstrap interval of 1.03–1.77 points.

The revised manuscript now presents prior published results only as contextual background and explicitly states that they are not directly comparable. All performance claims are restricted to the same-dataset, same-split evaluation above. A formal reimplementation of Zhou and Li remains a possible future analysis, but it would require the original preprocessing and candidate-space specifications; we do not substitute an unverified reproduction. This correction removes the unsupported cross-dataset claim while retaining a transparent, reproducible within-dataset comparison.

Revision in the manuscript — p. 4, Related Work

Manuscript text: "We do not claim that the model automates UX evaluation or improves usability; those outcomes require comparison with established UX methods and direct user-study evidence."

Comment 1.5 — Novelty

> *The novelty of the predictive modeling component should be clarified. The manuscript acknowledges prior work that already used LSTM- and Transformer- based models for predicting next UI interactions from click history, interface information, and time. The main technical models in the present manuscript also appear to be conventional LSTM and Transformer architectures. Therefore, the authors should more clearly distinguish whether the primary contribution is a new predictive architecture, a new dataset, a UX methodology, or the integration of existing prediction models into UX analysis. The novelty claims should be aligned with the actual methodological contribution.*

Response We agree that LSTM, GRU, and Transformer encoders are established components. We have therefore narrowed the modeling claim. The contribution is not a new primitive attention or recurrent operation, nor a claim that this is the first recurrent–attention hybrid. The proposed UNCP (User Next-Click Predictor) is a task-specific gated fusion architecture combined with a leakage-aware, user-disjoint evaluation protocol.

For an input prefix x_1:T, a shared embedding and positional encoding are passed through two parallel branches: h^G_T = GRU(E(x_1:T))_T and h^T_T = Transformer(E(x_1:T)+P)_T. The Transformer state is projected to the recurrent dimension and an elementwise gate is computed as g = sigmoid(W_g[h^G_T || h^T_T] + b_g). The fused representation is h_UNCP = LayerNorm(g ⊙ h^G_T + (1−g) ⊙ h^T_T), followed by dropout and a classifier over training-observed click labels. The gate is learned per example and hidden dimension, allowing the model to adapt the relative contribution of short-range recurrence and non-local attention.

Following this recommendation, we designed, implemented, and evaluated a family of task-specific UNCP variants rather than presenting the standard LSTM, GRU, or Transformer baselines as novel. The original UNCP combines parallel GRU and Transformer branches through a learned feature gate. UNCP 2 adds a training-only click-transition graph; UNCP 3 adds a Mamba-2-inspired structured state-space branch; UNCP 4 combines typed graph attention, a transition graph, candidate masking, GRU sequence encoding, and adaptive fusion; UNCP 5 combines UNCP 2 with a smoothed back-off transition model through a validation-selected adaptive fusion rule; and UNCP 6 uses dual GRU streams, run-length information, and a repeat-versus-switch prediction head. UNCP 5 is an adaptive probability-level ensemble, not a newly trained end-to-end neural architecture.

All variants were evaluated on the same repeat-inclusive, user-disjoint protocol. Mean Top-3 accuracy was 82.30% for UNCP, 83.58% for UNCP 2, 81.40% for UNCP 3, 81.08% for UNCP 4, 85.09% for UNCP 5 (SD 0.20 percentage points), and 84.08% for UNCP 6, compared with 79.91% for Markov-4. UNCP 5 therefore obtained the highest observed result on this dataset and protocol. We present this as dataset-specific empirical evidence, not universal superiority.

The revised novelty claim has three parts: (i) the task-specific UNCP model family and its gated, graph-aware, transition-fusion, and repeat-aware variants; (ii) evaluation on the repeat-inclusive enterprise clickstream; and (iii) a leakage-aware workflow with user-disjoint splitting, training-only preprocessing, frequency-stratified diagnostics, repeated seeds, and uncertainty reporting. The individual GRU, Transformer, graph, state-space, and transition components are established building blocks. Our methodological contribution is their specific integration and controlled empirical evaluation for next-click prediction, not a claim that the individual components or every hybrid combination are first-ever inventions.

Revision in the manuscript — pp. 12–13, Prediction Models

Manuscript text: “UNCP 5 combines the trained UNCP 2 probability vector with a smoothed back-off transition distribution over the last one to four observed clicks.”

Comment 1.6 — Future-event information leakage

> *Possible target-information leakage in the predictive formulation should be clarified. Equation 7 defines the next-click prediction as a function of \(s_{n+1}\), \(t_{n+1}\), and the previous click sequence. The manuscript should clearly explain whether the next page and next timestamp are available at inference time before the next click occurs. If \ (s_{n+1}\) or \(t_{n+1}\) are derived from the future event, their inclusion could introduce target leakage. A precise description of which information is available at prediction time is essential.*

Response We agree that the previous Equation 7 was ambiguous and could be read as giving the predictor access to the future event. We corrected the manuscript to:

\[

\hat{\mathbf p}_{n+1}=f_{\theta}(c_{\max(1,n-L+1):n}), \quad \hat c_{n+1}=\operatorname*{argmax}_{c}\hat p_{n+1}(c), \quad L=32

\]

Here, the input is the last 32 observed click labels (or the shorter available prefix), and the model returns a probability distribution over next-click labels. The actual next click is the evaluation target. The future destination page and timestamp are not inputs at inference time. Account ID is used only to form disjoint evaluation groups and is not a predictive feature.

The implementation was checked against the revised formulation: the primary models receive only padded click-prefix token IDs and their observed prefix lengths. No page field, future timestamp, or future event is passed to the model. Timestamps are used only for the separately reported chronological audit and split construction. Page labels are attached after prediction for stratification; destination-page analyses that use the future page are identified as future-page diagnostics and excluded from real-time performance claims.

This correction removes the target-information pathway implied by the old equation. We also updated the manuscript text and retained protocol tests that verify past-only contexts, training-only vocabularies, and padding invariance. The revised formulation now states exactly what is available at prediction time and separates predictive inputs from retrospective evaluation metadata.

Revision in the manuscript — p. 10, Prediction Task

Manuscript text: “Neither future clicks, user identity, page identity, nor timestamps are model inputs.”

Comment 1.7 — Implementation and reproducibility

> *Model implementation details are insufficient for reproducibility. The descriptions of the LSTM and Transformer explain the general architectures, but important experimental settings are not reported clearly enough. The manuscript should provide, at minimum, embedding dimensions, number of LSTM units/layers, Transformer layers and attention heads, dropout values, optimizer, learning rate, batch size, number of epochs, early-stopping strategy, maximum sequence length, padding/masking strategy, loss function, random seeds, and hyperparameter-selection procedure. The architectural descriptions currently remain largely conceptual.*

Response The new reproducible implementations use 32-dimensional embeddings; one 64-unit recurrent layer for LSTM/GRU; and a two-layer Transformer encoder with four attention heads, 64-dimensional ReLU feed-forward layers, sinusoidal positional encoding, residual connections, and post-residual layer normalization. Dropout is 0.2 before the final classifier and within the Transformer. The baseline recurrent and Transformer classifiers read the final real prefix token; the hybrid models add the branches described in the manuscript. Context length is capped at the last 32 observed clicks, and padding is on the right. Transformer padding is masked, and recurrent outputs are read at the final real token. Adam uses learning rate 0.003 (0.01 for logistic regression), weight decay 0.0001, gradient clipping at 1, batch size 256, and a maximum of 30 epochs. Early stopping uses validation cross-entropy with patience 5 and minimum improvement 0.0001; the best validation checkpoint is restored. No class weighting or page-based candidate masking is applied to the primary models. The configuration was fixed for this reanalysis rather than selected through test-set tuning; this is not an exhaustive architecture search. Unseen input labels use an unknown token; unseen target labels are scored incorrect. Logistic regression uses one-hot representations of the last three clicks and log-transformed prefix label counts. The execution environment records Python 3.12.0, PyTorch 2.11.0, and NumPy 2.2.5. These are new reference implementations, not a claim to have reproduced the original Keras architectures or their historical scores.

Revision in the manuscript — pp. 11–13, Prediction Models, and pp. 15–16, Data Preparation: Implementation and model selection

Manuscript text: "The LSTM and GRU baselines each have one recurrent layer with 64 units. The Transformer baseline has two encoder layers, four attention heads, 64-dimensional ReLU feed-forward layers, sinusoidal positional encodings, residual connections, and post-residual layer normalization." The new implementation paragraph also specifies context length, padding and masking, optimizer and learning rates, early stopping, seeds, vocabulary handling, and the limits of model selection.

Evidence: Experiment notebooks and model configuration.

Comment 1.8 — Repeated runs and uncertainty

> *Repeated runs and uncertainty estimates should be reported. Single accuracy values are presented for the models, but neural-network performance can vary depending on initialization and optimization. Please repeat the experiments over several random seeds and report mean values with standard deviations or confidence intervals. Statistical testing between the principal models would also strengthen the comparison.*

Response **The original fitted classifiers and UNCP 2 were evaluated across seeds 11, 23, and 37 on the fixed repeat-inclusive grouped-user split. UNCP 5 reuses the three saved UNCP 2 checkpoints; its**

three gate coefficients were selected on validation users only from a fixed 76-rule grid. Its held-out-user mean Top-1/Top-3 is 66.71%/85.09%, with sample seed SD 0.84/0.20 percentage points. The newly trained UNCP 6 uses the same three seeds and reaches 66.60%/84.08% Top-1/Top-3 (sample seed SD 0.51/0.22 percentage points). The per-seed predictions and training histories are retained, and all three variants are compared on identical held-out targets.

Uncertainty is reported at two levels. Seed SD describes variation across the three neural checkpoints conditional on this fixed split; UNCP 5 Top-3 has sample SD 0.20 percentage points. To compare UNCP 5 and UNCP 2, we averaged paired correctness across seeds on the same 10,833 targets and resampled the 11 held-out users as clusters. The Top-3 gain is 1.51 percentage points with an exploratory 95% cluster bootstrap interval of 1.03–1.77 points. This is a within-dataset estimate, not evidence of improved UX or external-app generalization. The earlier Holm-adjusted UNCP-versus-Markov-4 test refers only to that separate comparison.

The fixed split, three seeds, aggregate summary, per-seed training JSON files, prediction arrays, cluster intervals, and paired-comparison table are included in the reproducibility package. Because the held-out cohort contains only 11 user IDs and the seeds are not independent data samples, these intervals and tests are reported as conditional/exploratory uncertainty, not as population-level confirmation.

Revision in the manuscript — pp. 13 and 16, Evaluation Metrics / Table 1

Manuscript text: “Neural models are trained with seeds 11, 23, and 37; tables show the mean and sample standard deviation across these runs.”

Evidence: Per-seed model results and paired comparisons.

Comment 1.9 — Evidence for UX improvement

> *The claim that the approach improves user experience is currently stronger than the evidence presented. The manuscript describes several interface modifications and states that, after retesting with “some users,” interaction flows improved, sequence lengths decreased, and user flows became less complex. However, no sample size, control condition, pre/post quantitative results, statistical analysis, task-completion measures, or validated UX scales are reported. Consequently, the study presently provides stronger evidence for next-click prediction than for actual UX improvement. The latter claim should either be supported through a formal user study or reframed as a potential application.*

Response We agree with the reviewer’s assessment and withdraw the empirical claim that the approach has already improved user experience. The present evidence supports next-click prediction and clickstream-based decision support, not a causal UX effect. The earlier manuscript wording about retesting “some users,” shorter sequences, simpler flows, and improved interaction was not supported by a preserved sample size, control condition, pre/post measurements, task outcomes, validated UX scale, or statistical analysis. We have removed or reframed those statements throughout the abstract, UX implications, discussion, and conclusion. The formal UX-improvement claim is therefore deferred to future work.

The application was redesigned in the course of the project, but the available project time and resources were not sufficient to repeat the full user experiment and rerun the complete model evaluation on newly collected post-redesign data. A valid retest would require finalizing and redeploying the redesigned application, defining equivalent tasks, recruiting or identifying an appropriate participant sample, obtaining the required ethical and

organizational clearance, collecting a new controlled interaction log, and repeating the complete preprocessing and model pipeline. Those post-redesign data are not available in the current archive. We therefore do not present the redesign as a completed UX experiment and do not invent a sample size, control condition, effect, or significance result.

The revised claim is narrower and testable: knowing the predicted next click may help UX researchers identify which page regions and interface elements most strongly influence subsequent choices, locate recurrent transitions and unexpected deviations, prioritize screens for qualitative inspection, and formulate a redesign hypothesis. For example, an element that is repeatedly predicted or repeatedly followed by a particular transition can be considered when reviewing hierarchy, placement, feedback, or navigation. This is analytical decision support, not evidence that the redesign improves usability, reduces evaluation effort, or increases satisfaction.

To establish the latter, future work will require a sufficiently powered and ethically approved controlled pre/post or A/B study. We specify a preregistered sample, equivalent tasks, participant-level analysis, and a prespecified control condition. Primary outcomes should include task success, completion time, errors and backtracking, abandonment, time-to-click where appropriate, and validated usability and satisfaction scales, with confidence intervals and statistical tests. Next-click accuracy should be reported separately from these UX outcomes. Until this substantially larger follow-up study is completed, the manuscript makes no claim of demonstrated UX improvement; it presents the redesign only as a potential application of the predictive analysis and identifies formal UX validation as future work.

Revision in the manuscript — pp. 18–19, Proposed UX Improvements / Conclusion

Manuscript text: “Recorded outcomes from the stated retest are unavailable; therefore, this study does not claim that task success, satisfaction, sequence length, or usability improved.”

Comment 1.10 — Relationship to established UX methods

> *The relationship between next-click prediction and established UX testing methods requires stronger justification. The manuscript suggests that neural prediction can automate usability testing, first-click testing, and card sorting. These established UX methods measure more than click frequency or predicted next actions; they can involve user intent, information architecture, task success, cognition, satisfaction, and qualitative feedback. Predicting likely clicks may support these methods, but it should not automatically be treated as equivalent to conducting them. I recommend revising the terminology to describe the framework as an analytical or decision-support tool unless formal validation against conventional UX studies is provided.*

Response We agree. A next-click ranking is an analytical and decision-support signal, not an automated usability test. It does not measure user intent, whether a first click is correct for an assigned task, information architecture or conceptual grouping, task success, cognitive load, satisfaction, or qualitative feedback. We have revised the terminology throughout the manuscript: the framework may support UX researchers by identifying recurrent transitions, unexpected paths, low-confidence contexts, and interface elements worth inspecting, but it is not presented as equivalent to usability testing, first-click testing, or card sorting. Formal equivalence would require validation against conventional UX studies with task-based outcomes, participant-level analysis, validated scales, and qualitative evidence. No such validation was available in this study, so we make no equivalence claim.

Revision in the manuscript — p. 13, Potential Decision Support for UX Analysis — predicted rankings support later inspection; first-click studies and card sorting retain their distinct purposes.

Manuscript text: "These rankings may help UX researchers identify recurrent transitions, unexpected paths, and interface elements that merit inspection. They do not measure user intent, task success, cognitive load, satisfaction, or usability, and they are not equivalent to conventional UX testing."

Comment 1.11 — Session-threshold sensitivity

> *Sequence construction deserves a more rigorous sensitivity analysis. The manuscript uses two- and four-minute expiry thresholds and states that the difference did not significantly affect model results. Please provide the actual number and distribution of sequences produced under each threshold, corresponding model performance, and an appropriate statistical or sensitivity analysis. The rationale for choosing these thresholds should also rely primarily on peer-reviewed evidence rather than general web analytics resources.*

Response We agree that the earlier claim of no significant effect from the two- versus four-minute cutoff was unsupported, and we removed it from the revised manuscript. In the repeat-inclusive export, 4,185 rows containing 44,823 clicks have aligned timestamps. Splitting only within these rows at gaps greater than 120 and 240 seconds yields 4,306 and 4,263 sequences, respectively; the corresponding next-click target counts are 40,517 and 40,560. Mean sequence lengths are 10.41 and 10.51 clicks (median 5 and 95th percentile 25 for both settings). These are within-row refinements of the current dataset, not a reconstruction of missing source events.

Under the same grouped-account assignment, the first-order Markov Top-3 results are 65.07% and 65.10%, and fourth-order back-off Markov Top-3 results are 79.39% and 79.48%, for the 120- and 240-second variants, respectively (10,798 and 10,804 test targets). We report descriptive sensitivity results, not statistical equivalence or a causal effect of the cutoff, because the variants change the target sets. LSTM and Transformer were not rerun on these repeat-inclusive threshold variants; the earlier neural sensitivity results came from a smaller aligned-event reconstruction and are not used as results for the current primary dataset.

The two cutoffs are operational sensitivity settings, not universal behavioral constants or literature-established optimum values. The analysis excludes rows whose timestamps cannot be aligned and does not infer unobserved app-session boundaries. The manuscript now states this scope and gives the current counts and baseline performance; the detailed output files retain the complete sensitivity results.

Revision in the manuscript — pp. 6–7 and 13, Data Preprocessing / Evaluation Metrics

Manuscript text: "Within-row 120- and 240-second gap refinements and a collapsed-repeat variant assess preprocessing sensitivity. On the same grouped-account assignment, the first-order Markov Top-3 scores for the two gap settings are 65.07% and 65.10%, and the fourth-order back-off Markov scores are 79.39% and 79.48%, respectively."

Evidence: Session-threshold sensitivity outputs.

Comment 1.12 — Repeated-click filtering

> *Filtering repeated clicks may remove behavior that is important for UX research. Sequences containing repeated clicks on the same element were filtered as outlier behavior. Yet repeated clicking can represent slow response, unclear feedback, user confusion, accessibility difficulty, or interface frustration—all of which may be highly relevant to UX evaluation. The authors should justify this preprocessing decision and ideally evaluate whether retaining such events affects the results.*

Response We agree that repeated clicks can be meaningful UX signals. The revised primary analysis does not remove repeated clicks merely because the same element occurs consecutively. In the newly supplied raw export, we retained all 19,874 event rows and constructed a separate `add_repetitive` sequence dataset for all 50 raw user identifiers; adjacent repeated labels remain present. A sequence-level output and an event-level output are provided separately so that repetition can be audited without collapsing the raw observations. In the sorted raw export, 12,798 adjacent same-label transitions are observable under the supplied ordering.

For transparency, we also retain the earlier sensitivity analysis in which consecutive identical labels were collapsed. That variant reduced 35,290 click tokens to 33,645 and next-click examples from 31,074 to 29,429; retrained LSTM and Transformer Top-3 values were 78.75% (SD 0.62 pp) and 72.12% (SD 0.45 pp), respectively. Because collapsing changes the target population, this is a descriptive sensitivity comparison, not a causal estimate. The primary interpretation now preserves repetition, while the collapsed variant is reported only as a preprocessing sensitivity check.

Repeated clicks may reflect delayed response, missing feedback, confusion, accessibility difficulty, or frustration, but the click log alone cannot identify which explanation applies. We therefore do not label them as UX anomalies without corroborating latency, task, accessibility, or qualitative data. The raw and sequence-level `add_repetitive` outputs make these events available for future UX analysis and prevent the preprocessing step from silently discarding potentially meaningful behavior.

Revision in the manuscript — pp. 6–7, Data Preprocessing

Manuscript text: “Repeated clicks are retained in the revised primary analysis. The final analytic dataset contains 46,194 clicks, 4,223 sequences, 52 recorded User IDs, and 93 literal click labels.”

Evidence: Repeated-click reconciliation and sensitivity outputs.

Comment 1.13 — Population and ethics

> *The dataset population and ethical procedures should be described more fully. The dataset contains more than 35,000 clicks from 50 Android users interacting with an employee-recognition application. Because clickstream data are behavioral user data and User IDs were included in the dataset, the manuscript should state whether informed consent was obtained, whether the study received institutional/ethical approval or exemption, and how identifiers were anonymized or pseudonymized. The current manuscript should also discuss whether employees could reasonably have felt observed in a workplace context.*

Response Thank you for this important point. The platform supplied an opaque, UUID-formatted MX User Id. We consistently mapped each value to a numeric account code for sequence reconstruction and user-disjoint splitting; neither identifier was a predictive model input. This is pseudonymization, not proof of anonymity, and we cannot verify that the original UUID was cryptographically hashed or encrypted. The

analytical event table retains account code, click label, page, and time; the source export also contains Android advertising/device identifiers and IP addresses, which were excluded from the model features and analytical table. We have made this distinction explicit in the manuscript. Research-specific informed consent and prior ethics approval or exemption were not obtained. Because these are workplace behavioral records, employees may reasonably have felt observed; we have no evidence about their perception or freedom to decline. We therefore do not claim ethical clearance or participant anonymity. The Data Availability Statement points to the analytical files. Account-level source logs and the identifier mapping remain restricted pending institutional and editorial guidance; we removed an unsupported promise of access based on institutional authorization.

Revision in the manuscript — p. 20, Ethics and privacy statement — numeric account codes are described as pseudonymization, and absence of prior research consent and ethics approval is disclosed.

Manuscript text: "Replacing the User ID with a number is pseudonymization, not irreversible anonymization; we have no evidence that the original UUID was cryptographically hashed or encrypted. Research-specific informed consent and prior institutional ethics approval or exemption were not obtained."

Evidence: Ethics and privacy statement.

Comment 1.14 — Generalizability

> *The generalizability limitation should be expanded. The authors appropriately acknowledge limited user diversity and propose future expansion to additional users and app types. This limitation is substantial because the model is evaluated on a single enterprise application, Android users, and a relatively small user population. It is presently unclear whether the learned behavior transfers to other interfaces, user populations, devices, or application categories. An external-app or cross-user evaluation would considerably strengthen the contribution.*

Response We agree that generalizability remains limited. Within the available data and resources, we prioritized a stronger within-app test: all events from a recorded account were assigned to one partition, and the models were evaluated on 11 held-out User IDs. This assesses prediction for unseen accounts in the same Android enterprise application; it does not establish transfer to other organizations, interfaces, operating systems, or app categories. The archive contains 52 recorded User IDs, which have not been verified as 52 distinct people. We have made these boundaries explicit and do not claim wider portability. We used the available revision effort to correct the split, add baselines, and report uncertainty, but could not collect a larger, more diverse, multi-application cohort for this study. Future work should test new users across several application categories and device platforms under a fixed protocol, including an external-app or leave-one-app-out evaluation. We agree that these tests would materially strengthen the contribution.

Revision in the manuscript — p. 19, Conclusion and Future work — evaluation is limited to one Android enterprise application and 52 recorded User IDs.

Manuscript text: "Generalizability is limited by evaluation on one enterprise Android application and an uneven cohort of 52 recorded User IDs, which are not confirmed to represent 52 distinct people."

Comment 1.15 — Demonstrated findings versus future applications

> *The conclusions should distinguish demonstrated results from proposed future applications. The manuscript concludes that the models can significantly reduce the time and effort required for UX evaluation and enable scalable UX testing. Since formal A/B testing and established UX outcome metrics are explicitly deferred to future work, these claims should be moderated. The current results demonstrate next-click prediction performance; reductions in UX-testing time, improved usability, and increased user satisfaction require direct empirical validation.*

Response We agree with the reviewer and have applied this recommendation specifically in response to Reviewer 1. The manuscript no longer presents next-click prediction or element-frequency ranking as equivalent to first-click testing, usability testing, or card sorting. The demonstrated result is limited to next-click prediction under the reported user-disjoint protocol, with Top-1/Top-3 performance and uncertainty. Model outputs may serve as an analytical decision-support signal: they can help prioritize navigation paths, recurrent transitions, unexpected paths, and interface elements for later UX investigation. However, they do not measure user intent, first-click correctness for an assigned task, information architecture or conceptual grouping, task success, cognition, satisfaction, or qualitative feedback. We therefore removed claims that the models automate UX evaluation, replace established UX methods, reduce UX-testing time, or establish improved usability. Any methodological mapping to those methods is presented only as a future hypothesis; formal validation would require a controlled task-based study, participant-level analysis, validated UX scales, and qualitative evidence. No such equivalence or UX outcome is claimed from the present data.

Revision in the manuscript — pp. 18–19, Proposed UX Improvements / Conclusion — prediction findings are distinguished from unmeasured usability and satisfaction outcomes.

Manuscript text: "The study does not demonstrate a causal improvement in UX, reduced UX-evaluation time or effort, increased user satisfaction, or scalable replacement of first-click testing, card sorting, and controlled usability evaluation."

Comment 1.16 — Overall assessment

Response We sincerely thank the reviewer for this thoughtful and constructive assessment. The comments substantially improved our understanding of the methodological issues and helped us strengthen the manuscript. Within the available data and resources, we applied the recommendations as fully as possible: we replaced the overlapping split with a user-disjoint protocol, added transparent naïve and Markov baselines, restored repeated clicks, reported repeated-seed uncertainty and diagnostic analyses, clarified the model inputs and novelty, and moderated unsupported UX-improvement claims. We also implemented several UNCP variants that combine the sequence models discussed in the manuscript with gated fusion, transition graphs, graph attention, state-space-inspired processing, candidate masking, and repeat-aware components. On the same held-out-user evaluation, these models produced encouraging results; UNCP 5 achieved the highest observed mean Top-3 accuracy of 85.09%. We present this as a dataset- and protocol-specific result, not a universal superiority claim. The demonstrated contribution is next-click prediction and its possible use as decision support for UX analysis. Direct improvements in usability, satisfaction, or testing efficiency remain subjects for a future controlled study. We are grateful for the reviewer's valuable guidance and believe the resulting revision is clearer and scientifically stronger.

Revision in the manuscript — pp. 15–19, Results and Conclusion — the corrected user holdout, baselines, UNCP variants, and limited UX interpretation are summarized.

Manuscript text: “This study shows that next-click rankings can be learned from the enterprise application clickstream and used to prioritize interaction paths for UX inspection.”

Reviewer 2

Comment 2.1 — Methodological novelty

> *The methodological novelty is limited. The main predictive models are standard LSTM and Transformer architectures, and very similar next-click prediction approaches have already been studied in prior work, including the paper by Zhou and Li cited by the authors. The present contribution therefore appears mainly to be an application of established models to a new dataset rather than a new machine-learning approach. The manuscript should clarify what methodological advance is made beyond existing next- click prediction studies.*

Response Thank you for highlighting this limitation. We agree that LSTM, GRU, and Transformer are established architectures. In response, we focused the revision on designing, implementing, and evaluating a family of new task-specific models named UNCP rather than presenting the standard models themselves as novel. The original UNCP combines parallel GRU and Transformer branches through a learned gate. We then implemented UNCP 2–6, adding different combinations of a transition graph, graph attention, a Mamba-2-inspired state-space branch, candidate masking, adaptive transition fusion, and repeat-aware dual-GRU processing. All variants were evaluated on the same repeat-inclusive, user-disjoint protocol. Mean Top-3 accuracy ranges from 81.08% to 85.09%, with UNCP 5 obtaining the highest observed result of 85.09% (SD 0.20 percentage points), compared with 82.30% for the original UNCP and 79.91% for Markov-4. These results are encouraging for this dataset and protocol. We have clarified that the methodological contribution is the task-specific combination and empirical evaluation of these components, together with the leakage-aware protocol; we do not claim that LSTM, GRU, Transformer, graph operations, or transition models are individually new, or that the results establish universal superiority.

Revision in the manuscript — pp. 12–13 and 16, Prediction Models / Table 1

Manuscript text: "The individual operations are established building blocks; the contribution claimed here is their task-specific gated combination and its leakage-aware evaluation, not universal or first-ever architectural novelty."

Comment 2.2 — Generalization and user-level/temporal evaluation

> *The current train-test design does not convincingly demonstrate generalization. The dataset contains only 50 users, while User ID is explicitly included among the model inputs. The paper then applies a simple 80/20 split to 4,216 click sequences. If sequences from the same users appear in both training and test sets, the models may partly memorize user-specific behavior rather than generalize to unseen users. A user-level holdout and/or strictly temporal evaluation is necessary. This issue is particularly important given the small number of users.*

Response Thank you for identifying this important issue. We agree that the original random example-level split did not demonstrate generalization to unseen users. Our audit reconstructed the earlier procedure and found that 6,061 of 6,215 historical test examples (97.52%) shared their source sequence with training, while all 52 recorded User IDs appeared in both partitions. We therefore withdrew the old scores as evidence of unseen-user generalization. Following the reviewers' recommendations, we also restored 10,904 previously removed repeated clicks. The revised dataset increased from the earlier reported total to 46,194 click events and 4,223 sequences. The archive contains 52 recorded User IDs rather than the previously reported 50; because these IDs have not been reconciled with participant records, we report them as User IDs rather than

claiming 52 confirmed distinct people. We then reran the experiments using a user-disjoint protocol: complete User IDs were assigned to training, validation, and test sets before any prefix–target examples were generated. The 46,194 figure counts click events; it is not the number of supervised examples. Because the first click in each of the 4,223 sequences has no preceding click from which to form a target, the dataset yields 46,194 - 4,223 = 41,971 next-click targets. The resulting split contains 32/9/11 User IDs and 25,399/5,739/10,833 targets. These totals reconcile exactly to 52 User IDs and 41,971 targets. Automated overlap checks confirm that no User ID appears in more than one partition; because sequences and their prediction prefixes are generated only after account assignment, no source sequence can appear in both training and test sets. Vocabulary, transition statistics, and frequency estimates are fitted on training data only, while validation data control early stopping and model selection. User ID is used only to construct the groups and is not a predictive model input. All newly reported baselines and UNCP variants were executed under this corrected split. We also performed a separate qualified chronological evaluation on the valid-time subset, with the latest training event preceding validation and the latest validation event preceding testing. This temporal result is reported separately and is not presented as validation of timestamp-invalid portions of the archive.

Revision in the manuscript — pp. 13 and 15, Evaluation Metrics / Data Preparation — accounts are partitioned before prefix generation, with no account or source-sequence overlap.

Manuscript text: “Automated protocol checks confirmed that no User ID or source sequence occurs in more than one partition.”

Evidence: Split manifest and automated overlap checks.

Comment 2.3 — Same-data comparisons and uncertainty

> *The empirical comparison is not sufficiently rigorous. Results from prior studies appear to be compared directly with results obtained on the present dataset, although the datasets, user populations, interfaces, candidate click sets, and experimental protocols differ. Such cross-dataset accuracy comparisons cannot establish model superiority. Strong baselines should instead be implemented on the same data, including most- popular-click, page-conditioned popularity, Markov models, logistic regression, GRU/RNN, and other sequential baselines. Multiple random seeds, confidence intervals, and statistical comparisons should also be reported.*

Response We implemented and evaluated the requested same-data baselines on the identical repeat-inclusive, user-disjoint test partition. The comparison now includes uniform and popularity baselines, Markov orders 1--4, logistic regression, a vanilla Elman RNN, LSTM, GRU, Transformer, and UNCP 1--6. The main Top-3 results are 79.91% for Markov-4, 71.58% for logistic regression, 75.10% for the new RNN baseline, 78.00% for LSTM, 78.88% for GRU, 71.92% for Transformer, 81.08% for UNCP 4, 81.40% for UNCP 3, 82.30% for UNCP, 83.58% for UNCP 2, 84.08% for UNCP 6, and 85.09% for UNCP 5. The RNN result is the mean over seeds 11, 23, and 37 (Top-1 50.22% with seed SD 0.26 percentage points; Top-3 75.10% with seed SD 0.48 points). Its exploratory 95% held-out-user cluster interval for Top-3 is 71.85%--79.20%. Logistic regression reaches 44.24% Top-1 and 71.58% Top-3 on the same partition. All fitted neural baselines and UNCP variants use the same training-only vocabulary, validation-based checkpoint selection, and three fixed seeds. The manuscript tables now report these same-data results and uncertainty; cross-dataset numerical comparisons with prior publications are no longer used to claim superiority. Paired user-cluster intervals and sign-flip tests with Holm correction are reported for the prespecified principal comparisons, while the RNN is included as an additional transparent baseline with its seed variability and cluster interval. UNCP 5 remains

the highest observed point estimate, but this result is conditional on the dataset and protocol and is not presented as universal superiority.

Revision in the manuscript — pp. 15–17, Quantitative Evaluation Results / Tables 1–2 — logistic regression, RNN, GRU, LSTM, Transformer, Markov, and UNCP variants are compared.

Manuscript text: "The same held-out users and 10,833 prediction targets are used for every method in Table 1."

Evidence: Tables 1–2 and model-result files.

Comment 2.4 — Interface structure and Top-3 accuracy

> *The reported accuracy may be strongly affected by the structure of the interface. The application contains only 65 clickable elements across 15 pages, and some pages contain very few clickable candidates. Under such conditions, Top-3 accuracy can be naturally high, particularly on pages with only a small number of possible actions. Page-level candidate-set size, class imbalance, popularity baselines, and per-page performance should therefore be reported before interpreting an 88% Top-3 accuracy as strong predictive performance.*

Response We agree that the interface structure and candidate-set size must be reported before interpreting Top-3 accuracy. A supplementary page-inventory file has been added to the revision package as an Excel-compatible CSV attachment. It lists each application page and the corresponding number of clickable elements, making the candidate-space differences between pages explicit. We also added same-data baseline results, including uniform and popularity baselines, page-related diagnostic baselines where the available page labels permit them, Markov models, logistic regression, vanilla RNN, LSTM, GRU, and Transformer. These baselines were evaluated on the same repeat-inclusive, user-disjoint split with repeated neural runs and uncertainty estimates. The revised results are therefore interpreted relative to transparent baselines rather than the former cross-dataset 88% value. We added page-level performance, candidate-count summaries, label-frequency groups, and macro-averaged metrics to the analysis outputs and manuscript results. Because the supplied event export contains 19 recorded page strings and does not fully reconcile with the earlier 15-page/65-element application inventory, page-level results are labelled according to the supplied mapping. We will continue the page-level analysis using the attached inventory, reporting per-page candidate-set size, imbalance, baseline performance, and model performance; destination-page calculations remain retrospective diagnostics and are not model inputs.

Revision in the manuscript — pp. 13–17, Evaluation Metrics and Results

Manuscript text: "In addition to micro Top-1 and Top-3 accuracy, we report macro averages across held-out User IDs and target labels, results by recorded page and by training-label frequency group, and candidate-set diagnostics."

Comment 2.5 — UX outcomes

> *The claim that the method improves user experience is not empirically demonstrated. The manuscript describes several interface modifications based on click behavior and states that subsequent user flows became shorter and less complex. However, the paper does not report a formal experimental design, sample size for the retest, control condition, statistical test, task completion time, error rate, usability score, satisfaction measure, or other UX outcome. Indeed, the authors themselves state that formal A/B testing and UX metrics such as time-to-click and task success rate are left for future work. Therefore, the current evidence supports next-click prediction, but not the stronger conclusion that the proposed approach improves user experience.*

Response We agree that the evidence does not establish improved UX. The author supplying these materials has confirmed that recorded outcomes from the claimed retest are unavailable. We have removed statements that interaction flows improved, sequences shortened, UX testing time decreased, or satisfaction increased. The described interface changes are presented only as design hypotheses. We revised the title to “User Next Click Prediction Using Machine Learning for UX Analysis” to reflect the measured prediction task and its potential use in UX analysis. The abstract, contribution statement, discussion, and conclusion distinguish the prediction results from future UX outcome evaluation. No A/B study, participant sample, effect size or usability score has been invented. A controlled user study is required before making an empirical UX-improvement claim.

Revision in the manuscript — pp. 18–19, Proposed UX Improvements / Conclusion

Manuscript text: “These proposed improvements are informed by observed click patterns. Recorded outcomes from the stated retest are unavailable; therefore, this study does not claim that task success, satisfaction, sequence length, or usability improved.”

Comment 2.6 — Prediction versus UX research methods

> *The connection between prediction and UX methods is conceptually underdeveloped. First-click testing, usability testing, and card sorting are established UX research methods with specific experimental purposes. Simply predicting clicks or ranking frequently clicked elements is not equivalent to conducting these tests. The manuscript should either provide a validated methodological mapping between model outputs and each UX method or substantially moderate these claims.*

Response We have substantially moderated this claim. A predicted click ranking may guide which navigation paths to inspect or which design hypotheses to test, but it does not measure intent, correctness of the first click for an assigned task, conceptual grouping in card sorting, task success, satisfaction, cognition, or qualitative feedback. The revised UX section describes these boundaries and the outcomes a future controlled study would need. We make no claim of validated equivalence to first-click testing, card sorting or usability testing.

Revision in the manuscript — p. 13, Potential Decision Support for UX Analysis

Manuscript text: “They do not measure user intent, task success, cognitive load, satisfaction, or usability, and they are not equivalent to conventional UX testing.”

Comment 2.7 — Implementation detail

> *Important implementation details are missing. The manuscript does not sufficiently specify model hyperparameters, sequence padding/truncation, vocabulary construction, embedding dimensions, optimizer, learning rate, batch size, number of epochs, early stopping, class weighting, candidate masking, or model-selection procedure. These details are essential for reproducibility. The mathematical descriptions of the LSTM and Transformer are also very simplified and, in places, do not fully specify the actual computation used in the experiments.*

Response The new reproducible implementations use 32-dimensional embeddings; one 64-unit recurrent layer for LSTM/GRU; and a two-layer Transformer encoder with four attention heads, 64-dimensional ReLU feed-forward layers, sinusoidal positional encoding, residual connections, and post-residual layer normalization. Dropout is 0.2 before the final classifier and within the Transformer. All neural classifiers read the final real prefix token. Context length is capped at the last 32 observed clicks, and padding is on the right. Transformer padding is masked, and recurrent outputs are read at the final real token. Adam uses learning rate 0.003 (0.01 for logistic regression), weight decay 0.0001, gradient clipping at 1, batch size 256, and a maximum of 30 epochs. Early stopping uses validation cross-entropy with patience 5 and minimum improvement 0.0001; the best validation checkpoint is restored. No class weighting or page-based candidate masking is applied to the primary models. The configuration was fixed for this reanalysis rather than selected through test-set tuning; this is not an exhaustive architecture search. Unseen input labels use an unknown token; unseen target labels are scored incorrect. Logistic regression uses one-hot representations of the last three clicks and log-transformed prefix label counts. The execution environment records Python 3.12.0, PyTorch 2.11.0, and NumPy 2.2.5. These are new reference implementations, not a claim to have reproduced the original Keras architectures or their historical scores. We also correct the historical description: the archived attention code omitted the usual encoder feed-forward and normalization sublayers, so it should not have been described as if it implemented every standard Transformer computation.

Revision in the manuscript — pp. 11–13, Prediction Models

Manuscript text: “The vocabulary, popularity counts, transition frequencies, and smoothing priors are derived only from training data. Validation data select checkpoints and the UNCP 5 fusion rule; the test partition is evaluated after those choices are fixed.”

Evidence: Experiment notebooks and model configuration.

Comment 2.8 — Preprocessing validity

> *The dataset and preprocessing raise additional validity questions. Repeated clicks and other behaviors identified as “outliers” were removed, but these behaviors may themselves represent important usability problems. Similarly, the two- and four-minute session thresholds are only loosely motivated. Sensitivity analyses showing how these preprocessing decisions affect the conclusions would be useful.*

Response We sincerely thank the reviewer and the other reviewers for identifying the value of repeated-click behavior. We retain repeated clicks in the primary analysis and no longer classify repetition itself as an outlier. A repeated click may reflect delayed response, unclear feedback, confusion, accessibility difficulty, or interface frustration; removing it could therefore discard behavior that is directly relevant to UX diagnosis. By reconciling the pre-filter and post-filter exports, we restored 10,904 consecutive repeated-click events at their recorded user IDs and timestamps. The

repeat-inclusive primary dataset now contains 46,194 click events and 4,223 sequences, compared with 35,290 click events and 4,216 archived sequences before restoration—an increase of 10,904 events and 7 sequences (the spreadsheet row counts are 46,195 and 4,224 including the header). We also retained the original user identity and event order when reinserting each event. On the same user-disjoint protocol, mean Top-3 is 78.00% for LSTM, 71.92% for Transformer, 79.91% for Markov-4, 82.30% for UNCP, 83.58% for UNCP 2, 84.08% for UNCP 6, and 85.09% for UNCP 5. The repeat-aware UNCP 6 analysis stratifies 1,541 repeat targets and 9,292 switch targets: its Top-3 is 94.09%/82.42%, versus 94.07%/83.60% for UNCP 5, respectively. Because restoring repetitions changes the target population, these results are reported as the revised primary analysis rather than as a causal effect of preprocessing. A collapsed-repeat variant is retained only as a sensitivity analysis, not as the primary dataset. We removed the unsupported assertion that the two- and four-minute thresholds had no significant effect. On the current repeat-inclusive dataset, 4,185 timestamp-aligned rows contain 44,823 clicks. A within-row split at gaps greater than 120 or 240 seconds yields 4,306 or 4,263 sequences and 40,517 or 40,560 next-click targets, respectively. On the fixed grouped-account assignment, fourth-order back-off Markov Top-3 is 79.39% or 79.48% (10,798 or 10,804 test targets). These comparisons are descriptive because target sets differ. LSTM and Transformer were not rerun on these current threshold variants; earlier neural sensitivity results belong to a smaller aligned-event reconstruction and are not attributed to this dataset.

Revision in the manuscript — pp. 6–7 and 13, Data Preprocessing / Evaluation Metrics

Manuscript text: “Repeated clicks are retained in the revised primary analysis. The final analytic dataset contains 46,194 clicks, 4,223 sequences, 52 recorded User IDs, and 93 literal click labels.”

Evidence: Repeated-click and session sensitivity outputs.

Reviewer 4

Comment 4.1 — Numerical precision

> *Manuscript reports 88% Top-3 accuracy in the abstract, while Figure 10(b) shows the LSTM result as approximately 0.8891 and sec 4.6 describes it as nearly 89%. Use consistent precision.*

Response We agree and have corrected the headline precision. The historical 0.8891 LSTM result came from the contaminated overlapping split and is no longer used as the paper's primary result. The revised user-disjoint evaluation reports UNCP 5 at 85.09% mean Top-3 accuracy over three seeds; the abstract and prose report this consistently as 85% when rounded to the nearest whole percentage point, while tables report 85.09% ± 0.20 percentage points. The LSTM, Transformer, and other model values are likewise taken from the revised user-disjoint analysis. Figure 10(b), Section 4.6, the abstract, and the results tables now use the revised values and consistent precision. The former 88%/89% wording has been removed rather than relabelled.

Revision in the manuscript — pp. 1 and 16, Abstract / Table 1 — mean Top-3 is 85.09% for UNCP 5; the historical LSTM score appears only as protocol context (p. 15).

Manuscript text: "Mean Top-3 accuracy over three seeds is 85.09% for UNCP 5, 78.00% for LSTM, and 71.92% for Transformer."

Comment 4.2 — Training/test user overlap

> *Dataset is divided into 80% training and 20% testing, giving 843 test sequences from 4,216 sequences. But it is not explained whether sequences from the same user can appear in both sets.*

Response The original implementation did not split 4,216 sessions into 3,373/843 independent sessions. It expanded them into 31,074 overlapping prefix–target examples and randomly allocated 24,859/6,215 examples to training/test. A reconstruction of that procedure found 6,061 of 6,215 test examples (97.52%) shared their source session with training; all 52 recorded user IDs appeared in both partitions. We have corrected this description and withdrawn the old scores as evidence of unseen-user generalization. The new primary split assigns complete users to training/validation/test (32/9/11 user IDs; 25,399/5,739/10,833 next-click targets) with split seed 20261103. The click-event and target counts are different quantities: 46,194 events across 4,223 sequences yield 46,194 - 4,223 = 41,971 targets, and the three target partitions sum exactly to 41,971. Prefix generation follows user assignment, and vocabulary and frequency estimates use training data only. Model seeds 11, 23, and 37 are evaluated on the same held-out users. User ID is used for splitting only; contrary to the previous prose, inspection confirms the original notebooks also used only click labels as predictive input.

Revision in the manuscript — pp. 13 and 15, Evaluation Metrics / Data Preparation

Manuscript text: "The 46,194 figure counts click events, whereas the split counts are supervised next-click targets. A sequence of length L produces L-1 targets; therefore, 46,194 events across 4,223 sequences produce 41,971 targets. The fixed split contains 32, 9, and 11 User IDs and yields 25,399 training, 5,739 validation, and 10,833 test targets; these totals reconcile exactly to 52 User IDs and 41,971 targets."

Evidence: Split manifest and automated overlap checks.

Comment 4.3 — Model settings

> *Important training details for the LSTM and Transformer models are missing, including embedding dimension, LSTM units, number of Transformer layers and attention heads.*

Response The new reproducible implementations use 32-dimensional embeddings; one 64-unit recurrent layer for LSTM/GRU; and a two-layer Transformer encoder with four attention heads, 64-dimensional ReLU feed-forward layers, sinusoidal positional encoding, residual connections, and post-residual layer normalization. Dropout is 0.2 before the final classifier and within the Transformer. All neural classifiers read the final real prefix token. Context length is capped at the last 32 observed clicks, and padding is on the right. Transformer padding is masked, and recurrent outputs are read at the final real token. Adam uses learning rate 0.003 (0.01 for logistic regression), weight decay 0.0001, gradient clipping at 1, batch size 256, and a maximum of 30 epochs. Early stopping uses validation cross-entropy with patience 5 and minimum improvement 0.0001; the best validation checkpoint is restored. No class weighting or page-based candidate masking is applied to the primary models. The configuration was fixed for this reanalysis rather than selected through test-set tuning; this is not an exhaustive architecture search. Unseen input labels use an unknown token; unseen target labels are scored incorrect. Logistic regression uses one-hot representations of the last three clicks and log-transformed prefix label counts. The execution environment records Python 3.12.0, PyTorch 2.11.0, and NumPy 2.2.5. These are new reference implementations, not a claim to have reproduced the original Keras architectures or their historical scores.

Revision in the manuscript — pp. 11–13, Prediction Models

Manuscript text: "A shared 32-dimensional click embedding feeds two parallel branches: a 64-unit GRU for short transition patterns and a two-layer, four-head Transformer for longer contextual dependencies."

Evidence: Model configuration and experiment notebooks.

Comment 4.4 — Two/four-minute expiry sensitivity

> *Manuscript uses both 2-minute and 4-minute expiry times to create click sequences but only states that the model results were "not significantly affected." Any additional information will make this study more robust.*

Response We agree that the earlier claim of no significant effect from the two- versus four-minute cutoff was unsupported, and we removed it from the revised manuscript. In the repeat-inclusive export, 4,185 rows containing 44,823 clicks have aligned timestamps. Splitting only within these rows at gaps greater than 120 and 240 seconds yields 4,306 and 4,263 sequences, respectively; the corresponding next-click target counts are 40,517 and 40,560. Mean sequence lengths are 10.41 and 10.51 clicks (median 5 and 95th percentile 25 for both settings). These are within-row refinements of the current dataset, not a reconstruction of missing source events. Under the same grouped-account assignment, the first-order Markov Top-3 results are 65.07% and 65.10%, and fourth-order back-off Markov Top-3 results are 79.39% and 79.48%, for the 120- and 240-second variants, respectively (10,798 and 10,804 test targets). We report descriptive sensitivity results, not statistical equivalence or a causal effect of the cutoff, because the variants change the target sets. LSTM and Transformer were not rerun on these repeat-inclusive threshold variants; the earlier neural sensitivity results came

from a smaller aligned-event reconstruction and are not used as results for the current primary dataset. The two cutoffs are operational sensitivity settings, not universal behavioral constants or literature-established optimum values. The analysis excludes rows whose timestamps cannot be aligned and does not infer unobserved app-session boundaries. The manuscript now states this scope and gives the current counts and baseline performance; the detailed output files retain the complete sensitivity results.

Revision in the manuscript — pp. 6–7 and 13, Data Preprocessing / Evaluation Metrics

Manuscript text: "Within-row 120- and 240-second gap refinements and a collapsed-repeat variant assess preprocessing sensitivity. On the same grouped-account assignment, the first-order Markov Top-3 scores for the two gap settings are 65.07% and 65.10%, and the fourth-order back-off Markov scores are 79.39% and 79.48%, respectively."

Evidence: Session-threshold sensitivity outputs.

Comment 4.5 — Suggested cloud-load-balancing citation

> *Authors may consider citing recent work "Dynamic Load Balancing in Cloud Computing using Hybrid Kookaburra-Pelican Optimization Algorithms." This work discusses intelligent optimization for improving system performance and user experience and will provide broader supporting context under sec 2.*

Response We sincerely thank the reviewer for this valuable suggestion. We have added Addula et al.'s 2024 work, "Dynamic Load Balancing in Cloud Computing using Hybrid Kookaburra--Pelican Optimization Algorithms," to Section 2 (Related Work) and to the reference list. We cite it as broader context on hybrid intelligent optimization and adaptive system decisions. Because it addresses cloud-resource load balancing rather than next-click prediction or UX measurement, we do not use it as evidence for predictive accuracy, usability improvement, or equivalence to UX testing.

Revision in the manuscript — p. 4, Related Work

Manuscript text: "Addula et al. proposed a hybrid Kookaburra–Pelican optimization approach for dynamic cloud load balancing, targeting improved workload distribution and resource utilization."

Comment 4.6 — Repeated-click rule and removal count

> *Paper states that repeated clicks on same element were treated as outlier behavior and removed. Authors should explain rule used to identify these outliers and how many click events or sequences were removed during this preprocessing step, under sec 3.1.*

Response We sincerely thank the reviewer and the other reviewers for identifying the value of repeated-click behavior. We retain repeated clicks in the primary analysis and no longer classify repetition itself as an outlier. A repeated click may reflect delayed response, unclear feedback, confusion, accessibility difficulty, or interface frustration; removing it could therefore discard behavior that is directly relevant to UX diagnosis. By reconciling the pre-filter and post-filter exports, we restored 10,904 consecutive repeated-click events at their recorded user IDs and timestamps. The

repeat-inclusive primary dataset now contains 46,194 click events and 4,223 sequences, compared with 35,290 click events and 4,216 archived sequences before restoration—an increase of 10,904 events and 7 sequences (the spreadsheet row counts are 46,195 and 4,224 including the header). We also retained the original user identity and event order when reinserting each event. On the same user-disjoint protocol, mean Top-3 is 78.00% for LSTM, 71.92% for Transformer, 79.91% for Markov-4, 82.30% for UNCP, 83.58% for UNCP 2, 84.08% for UNCP 6, and 85.09% for UNCP 5. The repeat-aware UNCP 6 analysis stratifies 1,541 repeat targets and 9,292 switch targets: its Top-3 is 94.09%/82.42%, versus 94.07%/83.60% for UNCP 5, respectively. Because restoring repetitions changes the target population, these results are reported as the revised primary analysis rather than as a causal effect of preprocessing. A collapsed-repeat variant is retained only as a sensitivity analysis, not as the primary dataset.

Revision in the manuscript — pp. 6–7, Data Preprocessing

Manuscript text: "Repeated clicks are retained in the revised primary analysis. The final analytic dataset contains 46,194 clicks, 4,223 sequences, 52 recorded User IDs, and 93 literal click labels."

Evidence: Repeated-click reconciliation outputs.

Comment 4.7 — Suggested human-centered workload citation

> *Authors may consider citing the recent work "Human-centered AI for Personalized Workload Management: A Multimodal Approach to Preventing Employee Burnout." Its discussion of human-centered AI and personalization is relevant to the use of AI for improving user experience and can strengthen the background under sec 2.*

Response We sincerely thank the reviewer for this valuable suggestion. We have added Meduri et al. (2024), "Human-centered AI for personalized workload management: A multimodal approach to preventing employee burnout," to Section 2 (Related Work) and to the reference list (DOI: 10.24294/jipd.v8i9.6918). We use it as contextual support for human-centered AI, personalization, training, and feedback in workplace systems. Because the study addresses workload management and burnout rather than next-click prediction, we do not use it as evidence for predictive accuracy, UX improvement, or validation of first-click testing, card sorting, or usability testing. The revised manuscript also keeps the workplace and human-subject limitations explicit.

Revision in the manuscript — p. 4, Related Work

Manuscript text: "Meduri et al. studied how personalization, employee training, and feedback shape the effects of AI-supported workload management and emphasized that human-centered implementation matters beyond model performance."